U.G.C. NET Ex
COMPUTE
(Solved with Expl

1. The postfix expression AB + CD–* can be evaluated using a

(a) Stack (b) Tree

(c) Queue (d) Linked list

Ans. (a) : Postfix expression can be evaluated using operand stack.

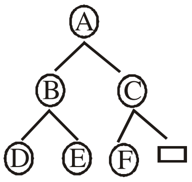
2. The post order traversal of a binary tree is DEBFCA. Find out the preorder traversal.

(a) ABFCDE (b) ADBFEC

(c) ABDECF (d) None of the above

Ans. (c) ABDECF – Because

By post order traversal, we cannot draw unique binary tree even with preorder and post order cannot define unique binary tree.



⇒ After considering this chart or binary tree, the entire traversal of the tree would be ABDECF. So the answer is option c.

Type	First	Second	Third
in order	L-Left	R_O-Root	R-Right
pre order	R_O-Root	L-Left	R-Right
post order	L-Left	R-Right	R_O-Root

3. The branch logic that provides making capabilities in the control unit is known as

(a) Controlled transfer

(b) Conditional transfer

(c) Unconditional transfer

(d) None of the above

Ans. (a) : Controlled transfer–that provider making capabilities in the control unit is unknown as control transfer.

4. The number of colours required to properly colour the vertices of every planer graph is

(a) 2 (b) 3

(c) 4 (d) 5

Ans. (a) : According to the 4-color theorem states that the vertices of every planar graph can be colored with at most 4 colors so that no two adjacent vertices receive the same color.

5. Networks that use different technologies can be connected by using

(a) Packets (b) Switches
(c) Bridges (d) Routers

Ans. (d): Routers–is a networking device that forwards data packets between computer networks. Router performs the traffic directing functions on the internet. A data packet is typically forwarded from one router to another through the networks that consitude their internetwork until it reaches its destination node. Router works at layer 3 of OSI model.

6. Both hosts and routers are TCP/IP protocol software. However, routers do not use protocol from all layers. The layer for which protocol software is not needed by a router is

(a) Layer–5 (Application)
(b) Layer–1 (Physical)
(c) Layer–3 (Internet)
(d) Layer–2 (Network Interface)

Ans. (a) : Layer–5 (Application)–Because Routing is a way to get one packet from one destination to the next. So (Application) layer–5 includes all the higher protocols. So the answer is Layer–5 (Application)

7. In multiuser database if two users wish to update the same record at the same time, they are prevented from doing so by

(a) Jamming (b) Password
(c) Documentation (d) Record lock

Ans. (d): Record lock– is the technique of preventing simultaneous access to data in a database, to prevent inconsistent results. If a collision is occure, both of the senders will send a jamming signal over the Ethernet.

8. A binary search tree is a binary tree

(a) All items in the left subtree are less than root
(b) All items in the right subtree are greater than or equal to the root
(c) Each subtree is itself a binary search tree
(d) All of the above

Ans. (d) : All of the above–The above mentioned properties are mentioned in book of Data Structures (DS). According to summary on binary search trees the following properties are

⇒ All items in the left subtree are less than root.
⇒ All items in the right subtree are greater than or equal to root.
⇒ Each subtree is itself a binary search tree (BST).

So the correct answer would be (d).

9. What deletes the entire file except the file structure?

(a) ERASE (b) DELETE

(c) ZAP (d) PACK

Ans. (C) : ZAP–Because ZAP Command removes all records from a table, leaving just a table structure. **ERASE** command deletes a file from disk.

10. Which command is the fastest among the following?

(a) COPY TO <NEW FILE>

(b) COPY STRUCTURE TO <NEW FILE>

(c) COPY FILE <FILE 1> <FILE 2>

(d) COPY TO MFILE-DAT DELIMITED

Ans. (b) : Copy structure to <new file> is just copying the structure only.

11. B + tree are preferred to binary tree in Database because

(a) Disk capacity are greater than memory capacities

(b) Disk access is much slower than memory access

(c) Disk data transfer rates are much less than memory data transfer rate

(d) Disks are more reliable than memory

Ans. (b) : Disk access is much slower than memory access. The major advantage of B^+ tree is in reducing the number of last level access which would be from disk in case of large data size, so B+ tree preferred option (b).

12. A Transaction Manager is which of the following?

(a) Maintains a log of transactions

(b) Maintains before and after database images

(c) Maintains appropriate concurrency control

(d) All of the above

Ans. (d) : All of the above

Because transaction manager performs all the operation mentioned in option A, B & C. So the answer is option (d).

13. Leaves of which of the following trees are at the same level?

(a) Binary tree (b) B–tree

(c) AVL–tree (d) Expression tree

Ans. (b) : A **B–tree** is a tree data structure in which each node has at most two children, which one referred as left child and the right child.

According to Knuth's definition, a B-tree of order m is a tree which satisfies the following properties:

Every node has at most m children

Every non-leaf node has at least m/2 children

The root has at least two children if it is not leaf node.

A non-leaf node with k children contains (k-1) keys.
All leaves appear in the same level.
It cannot be guaranteed in binary tree, AVL tree and expression tree.

14. Which of the following TCP/IP Internet protocol is diskless machine uses to obtain its IP address from a server?

(a) RAP (b) RIP
(c) ARP (d) X. 25

Ans. (c) : Because, ARP (Address Resolution Protocol) is a protocol in TCP/IP protocol suite. This protocol is used for Basic TCP/IP Operations. ARP is used to find the Ethernet (H/W) address from a specific IP number.

15. Decryption and encryption of data are the responsibility of which of the following layer?

(a) Physical layer (b) Data Link layer
(c) Presentation layer (d) Session layer

Ans. (c) : Service by **Presentation layer:**
Data conversion
Character code translation
Compression
Encryption and decryption

16. In which circuit switching, delivery of data is delayed because data must be stored and retrieved from RAM?

(a) Space division (b) Time division
(c) Virtual (d) packet

Ans. (b) : Time division
Because, Time division multiplexing (TDM) is a method of transmitting and receiving independent signals over a common signal Path by means of synchronized switches at each end of the transmission line so that each signal appears on the line only a fraction of time in alternating pattern.

17. In which Routing Method do all the routers have a common database?

(a) Distance vector (b) Link state
(c) Link vector (d) Dijkestra method

Ans. (b) : Link state necessary database require more memory than a distance vector requires.
Link state, the complex algorithm requires more CPU time than a distance vector protocol requires.
The flooding of link state packets adversely affects available bandwidth, particularly in unstable internetworks.

18. Page Shift Keying (PSK) Method is used to modulate digital signal at 9600 bps using 16 level. Find the line signals and speed (i.e. Modulation rate).

(a) 2400 bauds (b) 1200 bauds
(c) 4800 bauds (d) 9600 bauds

Ans. (a) : 2400 bauds

Modulation	Baud rate	Bit rate
4 – PSK	N	2N
8 – PSK	N	3N
16 – PSK	N	4N

Because, two values are provided, one is bps which is nothing but bits per second which is 9600 and also level which is 16. According to table given above for a 16–PSK give a baud rate N. the bit rate would be 4N. Here bps is provides which is 9600. In order to calculate the baud rate we have to divide bs/4. Because the signal level is 16. So the answer would be 9600/4 = 2400 baud.

19. The station to hub distance in which it is 2000 metres.

(a) 100 Base-Tx (b) 100 Base-Fx
(c) 100 Base-T_4 (d) 100 Base-T_1

Ans. (b) : 100 Base-Fx

Fast Ethernet is refered by 100 Base-x standard has three specification. Here 100 refers to the speed.

1. 100 Base T4
2. 100 Base Tx
3. 100 Base Fx

⇒ 100 Base Fx uses the 2-standard fibre optic cable and the station to hub distance is 2000 meters.

20. Main aim of software engineering is to produce

(a) Program
(b) Software
(c) Within budget
(d) Software within budget in the given schedule

Ans. (d) : Because every software would be produce within the specific budget and in the specific time.

21. Key process areas of CMM level 4 are also classified by a process which is

(a) CMM level 2 (b) CMM level 3
(c) CMM level 5 (d) All of the above

Ans. (c) : CMM level 5

Because, In CMM Model there are five maturity levels is identified by the number 1 to 5. They are–

(1) Initial, (2) Managed, (3) Defined, (4) Quantitavely Managed, (5) Optimizing

If a organisation is at level 2. It means it has crossed over level 1 and the same holds true and subsequent levels. At level 4, the process at level 3 is also included. CMM Level 5 is not correct because the question only talks about level 4. CMM level 2 is not correct because level 3 includes level & as well. So option (c) is correct.

22. **Validation means**

(a) Are we building the product right
(b) Are we building the right product
(c) Verification of fields
(d) None of the above

Ans. (b) : Are we building the right product
Because validation means "are we building the right product" and verification means "are we building the product right".

23. **If a process is under statistical control, then it is**

(a) Maintainable (b) Measurable
(c) Predictable (d) Verifiable

Ans. (c) : Predictable is a real valued stochastic process whose values are known in a sense, just in advance of time. Because, predictable process is also said to be under statistical control.

24. **In a function oriented design, we**

(a) Minimize cohesion and maximize coupling
(b) Maximize cohesion and minimize coupling
(c) Maximize cohesion and maximize coupling
(d) Minimize cohesion and minimize coupling

Ans. (b) : Maximize cohesion and minimize coupling.
Because, the design purpose is to making coupling minimize and maximize the cohesion. Cohesion is a way to understand that how close or bound your module is, and coupling is the level of interactivity between modules for a good design to happen. Cohesion should be more and coupling should be less.

25. **Which of the following metric does not depend on the programming language used?**

(a) Line of code (b) Function count
(c) Member of token (d) All of the above

Ans. (b) : Function count.
Because function count does not depend on programming language. Function count are a unit of measure a software just like a unit of measure for temperature would be degree.

26. **A/B + tree index is to be built on the name attribute of the relation STUDENT, Assume that all students names are of length 8 bytes, disk block are of size 512 bytes and index pointers are of size 4 bytes. Given this scenario what would be the best choice of the degree (i.e. number of pointers per node) of the B + tree?**

(a) 16 (b) 42
(c) 43 (d) 44

Ans. (c) : (43)

Because, Now we see how it comes,

Let n be the degree

given, key size (length of the name attribute of student) = 8 bytes (k), index pointer size = 4 bytes (b)

Disk Block Size = 512 bytes

Degree of B+ tree can be calculated if we know the maximum number of key a internal code can have. The formula is,

$\Rightarrow$ (n–1) k + n * b = block size

$\Rightarrow$ (n–1) * 8 + n * 4 = 512

$\Rightarrow$ 8n – 8 – 4n = 512

$\Rightarrow$ 12 = 520

$\Rightarrow$ n = 520/12 = 43 (Option c)

27. The In order traversal of the tree will yield a sorted listing of elements of tree in

(a) Binary tree (b) Binary search tree
(c) Heaps (d) None of the above

Ans. (b) : Binary search tree

Because, In BST all elements to the root of the tree will lesser than the root. Also elements greater than the root will be right of the root, so the answer is Option (b)

28. Mobile IP provides two basic functions

(a) Route discovery and registration
(b) Agent discovery and registration
(c) IP binding and registration
(d) None of the above

Ans. (b) : Agent discovery and registration

Because, Mobile IP is the basic behind how wireless devices offer IP connectivity.

Agent discovery: A mobile node discovers its foreign and home agents during discovery.

Registration: the mobile node registers its current location with the foreign agent and him agent during registration. Are two basic functions involved here, so the option is (b).

29. Pre-emptive scheduling is the strategy of temporarily suspending a gunning process

(a) Before the CPU time slice expires
(b) To allow starving processes to run
(c) When it requests I/O
(d) To avoid collision

Ans. (a) : Before the CPU time slice expires

Because, Every process is allocated a specific time slice in the CPU and it runs for that entire time. In pre-emptive scheduling even before the process time slice expires. It is temporarily suspended from its execution. So the option is (a).

30. In round robin CPU scheduling as time quantum is increased the average turn around time

(a) Increases (b) Decreases
(c) Remains constant (d) Varies irregularly

Ans. (d) : Varies irregularly

Because, ⇒ **Turn around time** is the interval of time between the submission of a process and its completion.

⇒ **Wait time** is the amount of time a process has been waiting in the ready queue

⇒ **Response time** is the time taken between the process submission and the first response produced.

In RR algorithm, value of time slice, plays a crucial role in deciding how effective the algorithm is. If quantum time is too small then it could be like context switching and if quantum time is high, the RR be haves like FCFS, the average response time varies irregularly. So the answer is option (d).

31. Resources are allocation to the process non-share below basis is :

(a) Mutual exclusion (b) Hold and wait
(c) No pre-emption (d) Circular wait

Ans. (a) : Mutual exclusion is a program object that prevents simultaneous access to a shared resource. This concept is used in concurrent programming with a critical section, a piece of code in which processes or threads access a shared resource.

32. Catch on and interleaved memories are ways of speeding up memory access between CPU and slower RAM. Which memory models are best suited (i.e. Improves the performance mostly which program)

(1) Cached memory is best suited for small loops.
(2) Interleaved memory is best suited for small loops
(3) Interleaved memory is best suited for large sequential code.
(4) Cached memory is best suited for large sequential code.

(a) 1 and 2 are true (b) 1 and 3 are true
(c) 4 and 2 are true (d) 4 and 3 are true

Ans. (b) : Interleaved memory is best suited for small 100 ps.

Because, Here multiple memory chips are grouped together to form what we are known as banks. Each of then take turns for supplying data. An interleaved memory with "n" banks is said to be n-way interleaved. Macintosh system are considered to be one using memory interleaving.

33. Consider the following page trace : 4, 3, 2, 1, 4, 3, 2, 1, 5 Percentage of page fault that would occur if FIFO page replacement algorithm is used with number of frame for JOB m = 4 will be :

(a) 8 (b) 9
(c) 10 (d) 12

Ans. (c) : 10
Because, reference string is 4, 3, 2, 1, 4, 3, 2, 1, 5
number of frames m = 4
⇒ First 4 references (4, 3, 2, 1) cause page faults and brought into empty frames
⇒ next reference (4) is already available and so there is no page fault
⇒ next reference (3) is also ready available and so there is no page fault
⇒ the next reference (5) replaces page 4 which was brought the first no. of page fault = 5
⇒ next reference (4) replace page 3 which is the next to come in no. of page faults = 6
⇒ next reference (3) replaces 2 no. of page faults till now = 7
⇒ next reference (2) replaces 1 a which was to last pages to come in no. of page fault till now = 8
⇒ next reference (1) replaces 5 which as the first to come in the second cycle no. of page faults till now = 9
⇒ the last reference in the reference string is 5 which will replace 4 no. of page faults till now = 10
⇒ the page faults are the 10 so the answer is option (c).

34. Check sum used along with each packet computes the sum of the data, where data is treated as a sequence of
(a) Integer (b) Character
(c) Real numbers (d) Bits

Ans. (b) : Character
Because, check sum is the error detecting mechanism when data is treated as a sequence of character. parity is a mechanism used when data is treated as a sequence of bits.

35. If an integer needs two bytes of storage, then the maximum value of a signed integer is
(a) $2^{16} - 1$ (b) $2^{15} - 1$
(c) 2^{16} (d) 2^{15}

Ans. (b) : $2^{15} - 1$
In case of magnitude Representation the Range is from $(2^{n-1}-1)$ to $(2^{n-1}-1)$
Min no. that can be represent in this system is $-(2^{n-1}-1)$
Max no. that can be represent in this system is $(2^{n-1}-1)$
In case 2's complement no. system the range is -2^{n-1} to $2^{n-1}-1$
Max no. that can be represent in this system $2^{n-1}-1$
So, Max no. can be represented here which $2^{n-1}-1$ is $2^{16}-1 \rightarrow 2^{15}-1$

36. Which of the following logic families is well suited for high-speed operations?
(a) TTI (b) ECL
(c) MOS (d) CMOS

Ans. (b) : ECL
Because, ECL (Emitter coupled logic) is a high speed integrated circuit bipolar transistor logic family.

37. **Interrupts which are initiated by an instruction are**

(a) Internal (b) External
(c) Hardware (d) Software

Ans. (c) : Hardware

Interrupts are of three types–(1) External interrupts, (2) Internal interrupts, (3) Software interrupts.

Hardware interrupts are used by devices to communicate that they require attention from the operating system.

38. **printf("%c" 100)**

(a) Prints 100
(b) Prints ASCII equivalent of 100
(c) Prints garbage
(d) None of the above

Ans. (b) : Prints ASCII equivalent of 100

Because the %c format prints the ASCII equivalent of the value.

39. **For the transmission of the signal, Bluetooth wireless technology uses**

(a) Time division multiplexing
(b) Frequency division multiplexing
(c) Time division duplex
(d) Frequency division duplex

Ans. (c) : Time division duplex is the application of time division multiplexing to separate outward and return signals.

Here Bluetooth technology uses time division duplex transmission duplex.

40. **Consider the following statements**

a. Recursive languages are closed under complementation.
b. Recursively enumerable languages are closed under union.
c. Recursively enumerable languages are closed under complementation.

Which of the above statements are true?

(a) 1 only (b) 1 and 2
(c) 1 and 3 (d) 2 and 3

Ans. (b) : 1 and 2

Because, recursive languages are closed under the following operations–

(1) Keene star, (2) concatenation, (3) union, (4) intersection, (5) complement, (6) set difference

Recursively enumerable languages are enclosed under the following operation.

(1) Keene star, (2) concatenation, (3) union, (4) intersection

Recursively enumerable languages are not closed under compliment, so the statement (1) and (2) are only true.

41. What is the routing algorithm used by RIP and IGRP?

(a) OSPF (b) Link-state

(c) Dynamic (d) Dijkstra vector

Ans. (d) : Dijkestra vector

Because, RIP (Routing information Protocol) and IGRP (interior Gateway Routing Protocol) are the example of distance Vector Routing Protocol and open shortest path (OSPF) is an example of Link State Routing Protocols. Distance vector algorithms are based on Belma and ford algorithm. Link State Routing Protocols is based on Dijkestra algorithm. So the options are OSPF, link state, dynamic are ruled out. So the answer is option (d).

42. Identify the incorrect statement

(a) The overall strategy drives the E-Commerce data warehousing strategy.

(b) Data warehousing in an E-Commerce environment should be done in a classical manner.

(c) E-Commerce opens up an entirely new world of web server.

(d) E-Commerce security threats can be grouped into three major categories.

Ans. (d) : The threat environment for E-Commerce data warehousing application, security threats can be grouped into three major categories.

Loss of data secrecy

Loss of data integrity

Loss of denial of service

Because, E-commerce security threats are more than 3 in number and so the incorrect statement is d.

43. Reliability of software is directly dependent on

(a) Quality of the design

(b) Number of errors present

(c) Software engineers experience

(d) User requirement

Ans. (b) : Software reliability is measured in term of Mean time between failures. Reliability of software is number between 0 and 1. Reliability increases when errors or bugs from the program are removed, so the option is b.

44. .... is not an E-Commerce application.

(a) House banking

(b) Buying stocks

(c) Conducting an auction

(d) Evaluating an employee

Ans. (d) : Evaluating an employee is not an E-commerce application.

45. **.................... is a satellite based tracking system that enables the determination of person's position.**

(a) Bluetooth
(b) WAP
(c) Short Message Service
(d) Global Positioning System

Ans. (d) : Global Positioning System is a satellite based navigation system.

46. **A complete microcomputer system consists of**

(a) Microprocessor
(b) Memory
(c) Peripheral equipment
(d) All of the above

Ans. (d) : All of the above
Because, In complete microcomputer system there is a Microprocessor or and a memory part as well and there is peripheral equipment part is also there, so the correct option is (d).

47. **Where does a computer add and compare data?**

(a) Hard disk (b) Floppy disk
(c) CUP chip (d) Memory chip

Ans. (c) : CUP chip
Because, Hard disk, Floppy disk, Memory chip are the storage devices so these are ruled out and now the correct option is (c).

48. **Pipelining strategy is called implement**

(a) Instruction execution
(b) Instruction prefetch
(c) Instruction decoding
(d) Instruction manipulation

Ans. (b) : Instruction prefetch is often combined with pipelining in an attempt to keep the pipeline busy.
So the option is (b).

49. **Which of the following data structure is linear type?**

(a) Strings (b) Lists
(c) Queues (d) All of the above

Ans. (d) : All of the above
Because, strings, Link lists and queues are linear type data structure because the data of these option were arranged or organized in sequential or linearly, where data elements attached one after another.

50. **To represent hierarchical relationship between elements, which data structure is suitable?**

(a) Dequeue (b) Priority
(c) Tree (d) All of the above

Ans. (c) : A Tree structure is a way of representing the hierarchical nature of a structure in a graphical form.
so the answer is option (c).